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SATISFACTION FOR n -TH ORDER LANGUAGES DEFINED IN n -TH ORDER LANGUAGES

WILLIAM CRAIG

Introduction. In this paper we show that for n -th order languages L' with p nonlogical constants, $2 \leq n < \omega$, $0 \leq p < \omega$, a notion of satisfaction can be defined in an n -th order language containing one additional nonlogical constant, say S . By the usual methods we also show that this notion cannot be defined in L' . Hence, in its present formulation, Beth's Theorem in [1] for first order languages has no analogue for L' .

Our defining expression is such that, given any values of the other nonlogical constants and any appropriate m -tuple, it allows us to determine whether or not the m -tuple belongs to the value S of S without considering the totality of objects which are of the same type as S . Whether every definition in an n -th order language is equivalent to one thus "predicative", and hence whether there is a formulation of Beth's Theorem which generalizes to higher orders, we do not know.

The falsehood for L' of the analogue of Beth's Theorem implies the falsehood of an analogue of an interpolation theorem for first order languages. The above definability of satisfaction for L' implies a result on finite axiomatizability in slightly richer languages. Details are given in § 4.

Our construction slightly modifies one by Vaught, sketched in § 7 of [2] (see also § 2 of [2]), where he notes that the semantical notions for L' , or perhaps more accurately certain modifications of these, can be introduced in an $(n+1)$ -th order language. He thus generalizes remarks by Tarski [5], who considers only languages for the natural numbers. Roughly speaking, Vaught adds to L' not only a symbol S for a binary relation of satisfaction but also first order symbols N , X , and V , using N and X to denote isomorphisms of the natural numbers under the successor relation and using V to denote mappings of the universe A onto the set $A^{(\omega)}$ of eventually constant sequences of elements of A . Although the value of his S is uniquely determined by the values of his N , X , V , the latter values are not uniquely determined. We therefore add three argument places to his relation of satisfaction, to "incorporate", as the first three terms of a 5-tuple belonging to the new relation, any values of N , X , V . If $2 \leq n < \omega$, then the new relation has the same order n as the original binary relation. Moreover, for certain purposes it serves as well.

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§ 1. Types, orders, languages. We now define functions τ which take sets A as arguments and which correspond to the finite types. For any set A , let $\mathbf{0}(A) = A$ and, for $0 < r < \omega$, let $(\tau_0, \dots, \tau_{r-1})(A) = \mathbf{P}(\{\langle X_0, \dots, X_{r-1} \rangle : X_0 \in \tau_0(A), \dots, X_{r-1} \in \tau_{r-1}(A)\})$, where $\mathbf{P}(Z) = \{Z' : Z' \subseteq Z\}$. We often write τA for $\tau(A)$.

Let $\tau_0, \dots, \tau_{r-1}$ be the *immediate predecessors* of $(\tau_0, \dots, \tau_{r-1})$, and let τ' be a *predecessor* of τ if and only if τ' is a member of each set which contains as members all immediate predecessors of τ and all immediate predecessors of each member. Let X be of *type* τ on A (or simply of type τ when A is determined by the context) if and only if $X \in \tau(A)$. Let X be of *order* n (on A), $n < \omega$, if and only if either X is of type $\mathbf{0}$ and $n = 0$ or, for some $\tau_0, \dots, \tau_{r-1}$ and $n_0, \dots, n_{r-1}$, X is of type $(\tau_0, \dots, \tau_{r-1})$, each element of type $\tau_0, \dots, \tau_{r-1}$ is of order $n_0, \dots, n_{r-1}$ respectively, and $n = 1 + \max(n_0, \dots, n_{r-1})$. Let X be *monadic* (on A) if and only if X is of a type τ such that τ and each of its predecessors has at most one predecessor. Thus, if we let $\mathbf{P}^0(Z) = Z$ and $\mathbf{P}^{n+1}(Z) = \mathbf{P}(\mathbf{P}^n(Z))$, then X is monadic and of order n on A if and only if $X \in \mathbf{P}^n(A)$.

We now form a language for a simple theory of types. Consider any τ . Our expression for it is either ' $\mathbf{0}$ ' or composed in a certain way from ' $\mathbf{0}$ ', commas, and parentheses. Let v_i^τ be the result of adding this expression to ' v ' as superscript and, say, the Arabic numeral for i as subscript, $i < \omega$, and let $v_0^\tau, \dots, v_i^\tau, \dots$ be the *variables of type* τ (in their *alphabetical* order). For any A , we intend $v_0^\tau, \dots, v_i^\tau, \dots$ to range over the objects of type τ on A . The *order* of v_i^τ , which we shall sometimes denote by $o(\tau)$, shall be the maximum of the lengths of nested sequences of parentheses in the superscript. Thus, for any A , the set of objects of order n is the union of the ranges of the variables of order n . Let v_i^τ be *monadic* if and only if τ and each of its predecessors has at most one immediate predecessor. Thus, if we let $(\mathbf{0})^0 = \mathbf{0}$ and $(\mathbf{0})^{n+1} = ((\mathbf{0})^n)$, then v_i^τ is monadic and of order n if and only if $\tau = (\mathbf{0})^n$. The *atomic formulas* shall be the expressions $v_{i_0}^{\tau_0} v_{i_1}^{\tau_1} \dots v_{i_r}^{\tau_r}$ where $\tau_0 = (\tau_1, \dots, \tau_r)$ and where here and elsewhere juxtaposition is used to indicate juxtaposition. The set of *formulas* shall be the least set of expressions which includes the set of atomic formulas and in which there are $\neg \psi$ and $\exists v_i^\tau \psi$ for any τ and i and for any ψ and ψ' in the set. Thus we have no function symbols, non-logical constants, identity symbol, or abstraction operators. Later on, however, certain variables will be treated as nonlogical constants. For $2 \leq n < \omega$, ψ shall be an L_n -*formula*, or simply $\psi \in L_n$, if and only if ψ is a formula such that any variable free (bound) in ψ is of order $\leq n$ ($< n$).

We let $v_j = v_j^0$, $j < \omega$. We shall often describe formulas by using in one of the usual ways some of the following eleven symbols: $\forall, \wedge, \vee, \neg, \supset, \equiv, \simeq, (,), [,]$. We shall use ' $\simeq$ ' only when the variables are of order 0. For example, $v_i^0 \simeq v_j^0$, $v_i \simeq v_j$, $\forall v_0^{(0)} [v_0^{(0)} v_i \equiv v_0^{(0)} v_j]$, and $\neg \exists v_0^{(0)} \neg [(v_0^{(0)} v_i \supset$

$v_0^{(0)}v_j \wedge (v_0^{(0)}v_j \supset v_0^{(0)}v_i)]$ are the same formula. We shall also sometimes use parentheses and commas in describing atomic formulas. For example, $v_2^{(0,0)}(v_7, v_3)$ is $v_2^{(0,0)}v_7^0v_3^0$.

Consider any formula ψ and any v_i^τ and v_j^τ . Let ψ' be the result of replacing in ψ the occurrences of v_j^τ , if any, within a quantifier having a free occurrence of v_i^τ in ψ within its scope, together with all occurrences of v_j^τ bound in ψ by these occurrences, by occurrences of v_k^τ , where v_k^τ is the first variable of type τ not occurring in ψ . Then letting $\psi = \psi(v_i^\tau)$ we shall mean by $\psi(v_j^\tau)$ the result of substituting v_j^τ for the free occurrences of v_i^τ in ψ' . For $\psi = \psi(v_{i_0}^{\tau_0}, \dots, v_{i_{r-1}}^{\tau_{r-1}})$ we define $\psi(v_{j_0}^{\tau_0}, \dots, v_{j_{r-1}}^{\tau_{r-1}})$ similarly, forming ψ' by replacing, if necessary, bound occurrences of $v_{j_0}^{\tau_0}, \dots, v_{j_{r-1}}^{\tau_{r-1}}$ in that order, and then substituting $v_{j_0}^{\tau_0}, \dots, v_{j_{r-1}}^{\tau_{r-1}}$ simultaneously for the free occurrences of $v_{i_0}^{\tau_0}, \dots, v_{i_{r-1}}^{\tau_{r-1}}$ respectively.

For convenience later on, we henceforth assume that the language just described has been arithmetized in one of the usual ways, i.e., that distinct expressions of the language have been mapped into distinct natural numbers, and thereby distinct sets or relations among expressions onto distinct sets or relations among numbers, in a manner which preserves conditions of effectiveness. Thus, for example, the set of formulas, each L_n , and, for each τ , the set of variables of type τ have been mapped onto recursive sets. All symbolism and terminology for notions in the language just described will henceforth be used for the corresponding arithmetic images.

§ 2. Formulas for mappings into sequences. We now elaborate part of Vaught's argument as outlined on p. 307 of [2]. Acquaintance with [2] will not be required. We always let $n, k, n_0, k_0, \dots$ be non-negative and $q, r, \dots$ positive integers. For any set B let B^ω be the set of those functions e whose domain is ω and whose range is a subset of B , i.e., the set of ω -termed sequences of elements of B . Let $B^{(\omega)}$ be the set of those $e \in B^\omega$ such that for some k_0 we have $e(k_0 + k) = e(k_0)$ for all k , i.e., the set of those sequences in B^ω which are eventually constant. Consider any infinite set A with a distinguished element a_0 and any h such that h maps A onto a set which includes $A^{(\omega)}$ and is included in A^ω . For any n and $\tau_0, \dots, \tau_{q-1}$ such that $n \geq o(\tau_0), \dots, n \geq o(\tau_{q-1})$ we are going to define a sequence, $\langle h_0, \dots, h_{q-1} \rangle$ such that: (i) Each h_i maps $\mathbf{P}^n(A)$ into $(\tau_i A)^\omega$; and (ii) For each $e_0 \in (\tau_0 A)^{(\omega)}, \dots, e_{q-1} \in (\tau_{q-1} A)^{(\omega)}$ there is some $Z \in \mathbf{P}^n(A)$ such that $e_0 = h_0(Z), \dots, e_{q-1} = h_{q-1}(Z)$. We call $\langle h_0, \dots, h_{q-1} \rangle$ the $[n; \tau_0, \dots, \tau_{q-1}]$ -sequence induced by a_0 and h . It will be used for our notion of satisfaction and also, in Lemma 2 below, for reducing the number of types of bound variables in formulas.

Suppose first that $n = 0$ and that $n \geq o(\tau_0), \dots, n \geq o(\tau_{q-1})$ so that $\tau_0 = \dots = \tau_{q-1} = \mathbf{0}$. Then for each $i < q$ let A be the domain of h_i and for each $z \in A$, let $h_i(z) = h(h(z)(i))$. Suppose now that $n > 0$, that

$n \geq o(\tau_0), \dots, n \geq o(\tau_{q-1})$, and that all $[n-1, \tau'_0, \dots, \tau'_{q'-1}]$ -sequences induced by a_0 and h have been defined, where $n > o(\tau'_0), \dots, n > o(\tau'_{q'-1})$. For $\tau_i = \mathbf{0}$ let $r_i = 1$ and for $\tau_i = (\tau''_0, \dots, \tau''_{r-1})$ let $r_i = r$. Let $s_0 = 1$ and let $s_i = i + 1 + r_0 + \dots + r_{i-1}$, $0 < i \leq q$. Let $\langle \tau'_0, \dots, \tau'_{q'-1} \rangle$ be the sequence of $q' = s_q - 1$ terms which is obtained from $\langle \tau_0, \dots, \tau_{q-1} \rangle$ by replacing simultaneously each term $\tau_i = \mathbf{0}$ by the two terms $\mathbf{0}, \mathbf{0}$ and each term $\tau_i = (\tau'_{s_i}, \tau'_{s_i+1}, \dots, \tau'_{s_i+r_i-1})$ having predecessors by the r_i+1 terms $\tau'_{s_i}, \tau'_{s_i}, \tau'_{s_i+1}, \dots, \tau'_{s_i+r_i-1}$ in the order indicated. Thus $\langle \tau'_0, \dots, \tau'_{q'-1} \rangle$ has for each $i < q$ the same τ'_{s_i} as s_i -th and as (s_i-1) -th term. Let $\langle h'_0, \dots, h'_{q'-1} \rangle$ be the $[n-1; \tau'_0, \dots, \tau'_{q'-1}]$ -sequence induced by a_0 and h . Let $\mathbf{P}^n(A)$ be the domain of each h_i , $i < q$. If τ_i has predecessors then, for each $Z \in \mathbf{P}^n(A)$, let $h_i(Z)$ be that e in $(\tau_i A)^\omega$ such that $e(k)$ is $\{ \langle W_0, \dots, W_{r_i-1} \rangle : \text{for some } Z' \in Z, (h'_{s_i-1}(Z'))(k) = (h'_{s_i}(Z'))(k) \text{ and } W_0 = (h'_{s_i}(Z'))(k), \dots, W_{r_i-1} = (h'_{s_i+r_i-1}(Z'))(k) \}$. If τ_i is $\mathbf{0}$ then, for each $Z \in \mathbf{P}^n(A)$ let $h_i(Z)$ be that e in A^ω such that, if $Z \neq \emptyset$ and if $(h'_{s_i}(Z'))(k) = a$ for each $Z' \in Z$, then $e(k) = a$, and if either $Z = \emptyset$ or $(h'_{s_i}(Z'))(k) \neq (h'_{s_i}(Z''))(k)$ for some $Z', Z'' \in Z$, then $e(k) = a_0$.

Evidently, each $[n, \tau_0, \dots, \tau_{q-1}]$ -sequence $\langle h_0, \dots, h_{q-1} \rangle$ induced by a_0 and h satisfies (i). Also it is easily seen that each $[0; \mathbf{0}, \dots, \mathbf{0}]$ -sequence $\langle h_0, \dots, h_{q-1} \rangle$ induced by a_0 and h satisfies (ii). Now consider any $[n; \tau_0, \dots, \tau_{q-1}]$ -sequence $\langle h_0, \dots, h_{q-1} \rangle$ induced by a_0 and h , where $n > 0$ and $n \geq o(\tau_0), \dots, n \geq o(\tau_{q-1})$. Let r_i, s_i, q', h'_j , and τ'_j be as above, $i < q, j < q'$. Consider any $e_0 \in (\tau_0 A)^\omega, \dots, e_{q-1} \in (\tau_{q-1} A)^\omega$. Let Z be the set of those $Z' \in \mathbf{P}^{n-1}(A)$ which for $e'_0 = h'_0(Z'), \dots, e'_{q'-1} = h'_{q'-1}(Z')$ satisfy: If $i < q$ and $\tau_i = (\tau'_{s_i}, \dots, \tau'_{s_i+r_i-1})$ then, for every k , either $e'_{s_i-1}(k) \neq e'_{s_i}(k)$ or $\langle e'_{s_i}(k), \dots, e'_{s_i+r_i-1}(k) \rangle \in e_i(k)$; and if $i < q$ and $\tau_i = \mathbf{0}$ then, for every k , $e'_{s_i}(k) = e_i(k)$. We want to show for each $i < q$ that $e_i = h_i(Z)$. Suppose first that $\tau_i = (\tau'_{s_i}, \dots, \tau'_{s_i+r_i-1})$. It is easily seen that, for each k , $(h_i(Z))(k) \subseteq e_i(k)$. Now consider any $\langle W_0, \dots, W_{r_i-1} \rangle \in e_i(k)$. There is a k_0 such that, for all m , $e_0(k_0+m) = e_0(k_0), \dots, e_{q-1}(k_0+m) = e_{q-1}(k_0)$. There also are $e'_0 \in (\tau'_0 A)^\omega, \dots, e'_{q'-1} \in (\tau'_{q'-1} A)^\omega$ such that, for all m , $e'_0(k_0+m) = e'_0(k_0), \dots, e'_{q'-1}(k_0+m) = e'_{q'-1}(k_0)$, such that $e'_{s_i-1}(k) = e'_{s_i}(k)$ and $W_0 = e'_{s_i}(k), \dots, W_{r_i-1} = e'_{s_i+r_i-1}(k)$, and such that for each $j < q$ and $m \leq k_0$: If τ_j has predecessors and if $e_j(m) \neq \emptyset$, then $e'_{s_j-1}(m) = e'_{s_j}(m)$ and $e'_{s_j}(m) = U_0, \dots, e'_{s_j+r_j-1}(m) = U_{r_j-1}$, for some $\langle U_0, \dots, U_{r_j-1} \rangle \in e_j(m)$; if τ_j has predecessors and if $e_j(m) = \emptyset$, then $e'_{s_j-1}(m) \neq e'_{s_j}(m)$; if $\tau_j = \mathbf{0}$, then $e'_{s_j}(m) = e_j(m)$. By the inductive hypothesis, there is some $Z' \in \mathbf{P}^{n-1}(A)$ such that $e'_0 = h'_0(Z'), \dots, e'_{q'-1} = h'_{q'-1}(Z')$. Then $Z' \in Z$. Hence $\langle W_0, \dots, W_{r_i-1} \rangle \in (h_i(Z))(k)$. Therefore $e_i(k) \subseteq (h_i(Z))(k)$. This concludes our argument that $e_i = h_i(Z)$. Suppose now that $\tau_i = \mathbf{0}$. If there is some τ_j having predecessors and some k such that $e_j(k) \neq \emptyset$, then some Z' is in Z , as we have just seen. If not, then a similar construction also yields some $Z' \in Z$. Hence $Z \neq \emptyset$. Moreover, for each $Z' \in Z$ and each k ,

$e'_{s_i}(k) = e_i(k)$. Hence $(h_i(Z))(k) = e_i(k)$ for each k . Therefore again $e_i = h_i(Z)$. This concludes the proof that each $[n; \tau_0, \dots, \tau_{q-1}]$ -sequence $\langle h_0, \dots, h_q \rangle$ induced by a_0 and h satisfies (i) and (ii).

We now construct L_n -formulas related to the definitions just given. Let u, z, w, u', z', w', u'' be $v_0, \dots, v_6$ respectively, and let $N = v_0^{(0)}$, $X = v_0^{(0,0)}$, and $V = v_0^{(0,0,0)}$. For each n let $Z^n = v_1^{(0)^n}$, so that Z^n is that monadic variable v_1^τ such that $o(\tau) = n$, and for each τ let $W^\tau = v_2^\tau$. Let λ_1 be a formula whose free variables are N and X , whose bound variables are monadic and of order ≤ 1 , and whose models¹ are those $\langle A, N, X \rangle$ such that $\langle N, X \rangle$ is isomorphic to $\langle \omega, Sc \rangle$ where Sc is the successor relation. Let $\Delta_0 = \Delta_0(u) = Nu \wedge \forall z \neg X(z, u)$, and for $n > 0$ let $\Delta_n = \Delta_n(u) = \exists z [\Delta_{n-1}(z) \wedge X(z, u)]$. Let λ_2 be a conjunction of the universal closures of the following three formulas: $N(u) \supset \exists w V(z, u, w)$; $V(z, u, w) \wedge V(z, u, w') \supset w \simeq w'$; $N(u) \supset \exists z' [V(z', u, w) \wedge \forall u' \forall w' [[\neg u' \simeq u \wedge V(z, u', w')] \supset V(z', u', w')]]$. Let $\lambda = \lambda_1 \wedge \lambda_2$.

Consider any model $\mathfrak{M} = \langle A, N, X, V \rangle$ of λ . Let f be the isomorphism of $\langle N, X \rangle$ onto $\langle \omega, Sc \rangle$. Then $V(z, u, w)$ *determines* in $\mathfrak{M}$ a unique mapping h of A into A^ω such that, for each $z \in A$ and $k < \omega$, $(h(z))(k) = w$ if and only if, for some u , $f(u) = k$ and $\langle z, u, w \rangle \in V$. More generally, consider any formula ϑ such that the only variables free in ϑ are N, X, V, Z^n, u, W^τ and such that for each $Z \in \mathbf{P}^n(A)$ and each $u \in N$ there is exactly one W (which must be of type τ on A) such that $(\mathfrak{M}, Z, u, W)$ is a model of ϑ . Then ϑ *determines* in $\mathfrak{M}$ a unique mapping h of $\mathbf{P}^n(A)$ into $(\tau A)^\omega$ such that for each $Z \in \mathbf{P}^n(A)$ and $k < \omega$, $(h(Z))(k) = W$ if and only if, for some u , $f(u) = k$ and $(\mathfrak{M}, Z, u, W)$ is a model of ϑ . For verbal convenience, we shall also say that Δ_0 *determines* a_0 in $\mathfrak{M}$ when a_0 is the unique u such that $(\mathfrak{M}, u)$ is a model of $\lambda \wedge \Delta_0$. Note that the h determined by $V(z, u, w)$ in $\mathfrak{M}$ maps A onto a set which includes $A^{(\omega)}$.

For each n and $\tau_0, \dots, \tau_{q-1}$ such that $n \geq o(\tau_0), \dots, n \geq o(\tau_{q-1})$ we are going to construct unique L_n -formulas $\vartheta_0, \dots, \vartheta_{q-1}$ of a special kind which under conditions described below determine $h_0, \dots, h_{q-1}$ respectively. We call $\langle \vartheta_0, \dots, \vartheta_{q-1} \rangle$ the $[n; \tau_0, \dots, \tau_{q-1}]$ -sequence of formulas. Suppose first that $n = 0$ and that $\tau_0 = \dots = \tau_{q-1} = \mathbf{0}$. Then let ϑ_i be $\exists z' [V(z', u, w) \wedge \exists u [\Delta_i \wedge V(z, u, z')]]$, $i < q$. Suppose now that $n > 0$, that $n \geq o(\tau_0), \dots, n \geq o(\tau_{q-1})$, that r_i, s_i, q' , and τ_j are as above, $i < q$, $j < q'$, and that the $[n-1; \tau'_0, \dots, \tau'_{q'-1}]$ -sequence of formulas is $\langle \vartheta'_0, \dots, \vartheta'_{q'-1} \rangle = \langle \vartheta'_0(W_{\tau_0}^{\tau_0}), \dots, \vartheta'_{q'-1}(W_{\tau'_{q'-1}}^{\tau'_{q'-1}}) \rangle$. If τ_i has predecessors, let $\tau' = \tau'_{s_i+j}$ and $W_j = v_{2+j}^{\tau'}$, $j < r_i$, and let ϑ_i be $\forall W_0 \dots \forall W_{r_i-1} [W^{\tau_i}(W_0, \dots, W_{r_i-1}) \equiv \exists Z^{n-1} [Z^n(Z^{n-1}) \wedge \vartheta'_{s_i-1} \wedge \vartheta'_{s_i}(W_0) \wedge \dots \wedge \vartheta'_{s_i+r_i-1}(W_{r_i-1})]]$. If τ_i is $\mathbf{0}$,

¹ A model shall always be standard; non-standard Henkin-type models will not be considered.

let ξ_i be $\exists Z^{n-1}[Z^n(Z^{n-1})] \wedge \forall Z^{n-1}[Z^n(Z^{n-1}) \supset \vartheta'_i]$ and let ϑ_i be $\xi_i \vee [\Delta_0(w) \wedge \forall w \neg \xi_i]$.

We now state two lemmas. The first is proved by induction on n .

LEMMA 1. *Let $n \geq o(\tau_0), \dots, n \geq o(\tau_{q-1})$ and let $\langle \vartheta_0, \dots, \vartheta_{q-1} \rangle$ be the $[n; \tau_0, \dots, \tau_{q-1}]$ -sequence of formulas. Then: (i) For any $i < q$, ϑ_i contains free no variables other than N, X, V, Z^n, u , and W^{τ_i} and bound no variables other than $Z^0, \dots, Z^{n-1}$ or of a type which is $\mathbf{0}$ or a predecessor of τ_i . (ii) For any model $\mathfrak{M}$ of λ , if a_0 and h are determined in $\mathfrak{M}$ by Δ_0 and $V(z, u, w)$, and if $\langle h_0, \dots, h_{q-1} \rangle$ is the $[n; \tau_0, \dots, \tau_{q-1}]$ -sequence induced by a_0 and h , then ϑ_i determines h_i in $\mathfrak{M}$, $i < q$.*

LEMMA 2. *There is a recursive function g such that if ψ is any formula then: (i) $g(\psi)$ is a formula (the "transform" of ψ) having the same infinite models; (ii) each variable free in $g(\psi)$ is also free in ψ ; and (iii) for each variable $v_j^{\tau_j}$ bound in $g(\psi)$, one of the following holds: (a) $v_j^{\tau_j}$ is of type $\mathbf{0}$, $(\mathbf{0})$, $(\mathbf{0}, \mathbf{0})$, or $(\mathbf{0}, \mathbf{0}, \mathbf{0})$; (b) $v_j^{\tau_j}$ is monadic and of order $\leq n(\psi)$, where $n(\psi)$ is the highest of the orders of the variables bound in ψ ; or (c) $v_j^{\tau_j}$ is of a type which is a predecessor of a type of a variable free in ψ .*

PROOF. Let ψ be any formula with a bound variable $v_j^{\tau_j}$ satisfying none of the conditions (a), (b), (c). Then $\tau \neq \mathbf{0}$, so that $\tau = (\tau_0, \dots, \tau_{q-1})$ for some $\tau_0, \dots, \tau_{q-1}$. We can choose τ such that if $o(\tau') > o(\tau) = n$ and if $v_k^{\tau'_k}$ occurs bound in ψ , then $v_k^{\tau'_k}$ satisfies (a), (b), or (c). We can also assume that all variables bound in ψ are distinct from N, X, V, u, Z^{n-1}, Z^n and that no two quantifier occurrences in ψ contain the same variable, since there is an effective procedure of obtaining for each formula ψ' not satisfying these conditions a formula ψ which does, such that ψ and ψ' have the same free variables, the same types of bound variables, and the same models. For the time being we assume in addition that all variables free in ψ are likewise distinct from u, Z^{n-1}, Z^n , and that N, X, V do not occur in ψ as arguments of other variables. Now there is no monadic $v_k^{\tau'_k}$ such that $v_j^{\tau_j}$ occurs in ψ as argument of $v_k^{\tau'_k}$, i.e., in the atomic formula $v_k^{\tau'_k}(v_j^{\tau_j})$, since otherwise $v_j^{\tau_j}$ would also be monadic and thus satisfy (b). Nor is there a nonmonadic $v_k^{\tau'_k}$ such that $v_j^{\tau_j}$ occurs in ψ as argument of $v_k^{\tau'_k}$. Otherwise, if $v_k^{\tau'_k}$ occurred free in ψ , then $v_j^{\tau_j}$ would satisfy (c), and if $v_k^{\tau'_k}$ occurred bound in ψ , this would contradict our choice of τ , since the failure of $v_j^{\tau_j}$ to satisfy (a), (b), or (c) implies the failure of $v_k^{\tau'_k}$ to satisfy (a), (b), or (c). Thus each atomic formula in ψ in which $v_j^{\tau_j}$ occurs is of the form $v_j^{\tau_j}(v_{i_0}^{\tau_0}, \dots, v_{i_{q-1}}^{\tau_{q-1}})$. By our conditions on ψ , there is a formula $\exists v_j^{\tau_j} \chi$ occurring as part of ψ such that all occurrences of $v_j^{\tau_j}$ in ψ are occurrences in this part and bound by the initial quantifier of the part. Let χ^* be the result of replacing in χ each atomic formula $v_j^{\tau_j}(v_{i_0}^{\tau_0}, \dots, v_{i_{q-1}}^{\tau_{q-1}})$ by $\exists u \exists Z^{n-1}[Z^n(Z^{n-1}) \wedge \Delta_0 \wedge \vartheta'_0(v_{i_0}^{\tau_0}) \wedge \dots \wedge \vartheta'_{q-1}(v_{i_{q-1}}^{\tau_{q-1}})]$, where $\langle \vartheta'_0(W^{\tau_0}), \dots, \vartheta'_{q-1}(W^{\tau_{q-1}}) \rangle$ is the $[n-1; \tau_0, \dots, \tau_{q-1}]$ -sequence of formulas. Let ψ^* be the result of replacing $\exists v_j^{\tau_j} \chi$ in ψ by $\exists Z^n \chi^*$. Using the comprehension axiom one easily shows that $\lambda \wedge \exists v_j^{\tau_j} \chi$

and $\lambda \wedge \exists Z^n \chi^*$ have the same models. Hence $\lambda \wedge \psi$ and $\lambda \wedge \psi^*$ have the same models. Moreover, fewer bound variables in ψ^* than in ψ satisfy none of (a), (b), (c). Thus there is an effective method, namely repetition of this process a sufficient number of times, whereby given any ψ not containing N, X, V we find some $g'(\psi)$ such that each variable bound in $g'(\psi)$ satisfies (a), (b), or (c), such that each variable free in $g'(\psi)$ is either N, X, V or free in ψ , and such that $\lambda \wedge \psi$ and $\lambda \wedge g'(\psi)$ have the same models. Let ψ be without occurrences of N, X, V , and let $g(\psi) = \exists N \exists X \exists V [\lambda \wedge g'(\psi)]$. Then $\psi \wedge \exists N \exists X \exists V \lambda$, $\exists N \exists X \exists V [\lambda \wedge \psi]$, and $g(\psi)$ have the same models. But also ψ and $\psi \wedge \exists N \exists X \exists V \lambda$ have the same infinite models. It follows that $g(\psi)$ satisfies (i), (ii), and (iii). In the case where N, X , or V occurs in ψ , we define $g(\psi)$ analogously choosing in place of N, X, V suitable variables of the same type. Q.E.D.

§ 3. Modified satisfaction implicitly defined. We now turn to those n -th order languages L' , $n \geq 2$, which correspond to applied languages with a finite number $p \geq 0$ of non-logical constants. More precisely, consider any $n \geq 2$ and any finite, possibly empty, set $\{R_0, \dots, R_{p-1}\}$ of p variables of order $\leq n$. For convenience we assume until further notice that $R_0, \dots, R_{p-1}$ are of order n . Let $\{\tau_0, \dots, \tau_{q-1}\}$ be the set of those τ which satisfy at least one of the following conditions: $\tau = (\mathbf{0}, \mathbf{0})$; $\tau = (\mathbf{0}, \mathbf{0}, \mathbf{0})$; $\tau = (\mathbf{0})^m$ for some $m < n$, so that the variables v_j^{τ} are monadic and of order $< n$; τ is a predecessor of the type of some R_j , $j < p$. We let $\tau_0, \dots, \tau_{q-1}$ be distinct and, for convenience, let $\mathbf{0} = \tau_0$. Let L' be the set of those ψ in L_n such that any variable free in ψ is either one of $R_0, \dots, R_{p-1}$ or of one of the types $\tau_0, \dots, \tau_{q-1}$. Let L'' be the set of those ψ in L' such that any variable bound in ψ is of one of the types $\tau_0, \dots, \tau_{q-1}$. Thus, if ψ is in L' and g is as in Lemma 2, then $g(\psi)$ is in L'' .

Let $R_0, \dots, R_{p-1}$ be of type $\sigma_0, \dots, \sigma_{p-1}$ respectively. Consider any $\mathfrak{A} = \langle A, R_0, \dots, R_{p-1} \rangle$ such that $R_j \in \sigma_j A$, $j < p$. For any $\psi \in L''$ and $e_0 \in (\tau_0 A)^\omega, \dots, e_{q-1} \in (\tau_{q-1} A)^\omega$, consider the usual relation of *satisfaction* of ψ in $\mathfrak{A}$ by $\langle e_0, \dots, e_{q-1} \rangle$, where $R_0, \dots, R_{p-1}$ are assigned the values $R_0, \dots, R_{p-1}$ respectively and thus treated as non-logical constants, and where each $v_j^{\tau_i}$ is assigned the value $e_i(j)$, $i < q$. Then $\langle e_0, \dots, e_{q-1} \rangle$ satisfies ψ in $\mathfrak{A}$ if and only if there are $e'_i \in (\tau_i A)^\omega$ such that $e'_i(j) = e_i(j)$ whenever $v_j^{\tau_i}$ is free in ψ , $i < q$, and such that $\langle e'_0, \dots, e'_{q-1} \rangle$ satisfies ψ in $\mathfrak{A}$.

Consider for a moment the case where A is infinite. Consider any 1-1 mapping f of a subset of A onto ω , any $a_0 \in A$, and any h mapping A onto a set including $A^{(\omega)}$ and included in A^ω . Let $\langle h_0, \dots, h_{q-1} \rangle$ be the $[n-1; \tau_0, \dots, \tau_{q-1}]$ -sequence induced by a_0 and h . Then the satisfaction relation (for L' in $\mathfrak{A}$) induced by f , a_0 , and h shall be the set $Sat_I(L', \mathfrak{A}, f, a_0, h)$ of

those $\langle Z, u \rangle$ such that $Z \in \mathbf{P}^{n-1}(A)$, $f(u) \in L'$, and $\langle h_0(Z), \dots, h_{q-1}(Z) \rangle$ satisfies $g(f(u)) \in L''$ in $\mathfrak{A}$ in the way described.

From this definition and Lemma 2 we conclude:

LEMMA 3. *If $\mathfrak{A}$, f , a_0 , h are as just described, and if ψ is an L' -formula in which no variable other than $R_0, \dots, R_{p-1}$ occurs free, then $\mathfrak{A}$ is a model of ψ if and only if for some (all) $Z \in \mathbf{P}^{n-1}(A)$, $\langle Z, f^{-1}(\psi) \rangle \in \text{Sat}_I(L', \mathfrak{A}, f, a_0, h)$.*

Recalling that the $[n-1; \tau_0, \dots, \tau_{q-1}]$ -sequence $\langle h_0, \dots, h_{q-1} \rangle$ induced by a_0 and h has the property that for each $e_0 \in (\tau_0 A)^{(\omega)}$, $\dots$, $e_{q-1} \in (\tau_{q-1} A)^{(\omega)}$ there is some $Z' \in \mathbf{P}^{n-1}(A)$ such that $e_0 = h_0(Z')$, $\dots$, $e_{q-1} = h_{q-1}(Z')$ we also conclude:

LEMMA 4. *If $\mathfrak{A}$, f , a_0 , h are just described, if $\langle h_0, \dots, h_{q-1} \rangle$ is the $[n-1; \tau_0, \dots, \tau_{q-1}]$ -sequence induced by a_0 and h , and if $\exists v_m^{\tau_i} \psi$ is an L'' -formula, then $\langle Z, f^{-1}(\exists v_m^{\tau_i} \psi) \rangle \in \text{Sat}_I(L', \mathfrak{A}, f, a_0, h)$ if and only if $\langle Z', f^{-1}(\psi) \rangle \in \text{Sat}_I(L', \mathfrak{A}, f, a_0, h)$ for some $Z' \in \mathbf{P}^{n-1}(A)$ satisfying these two conditions: (i) if $k \neq m$ and $v_k^{\tau_i}$ is free in ψ , then $(h_i(Z))(k) = (h_i(Z'))(k)$; and (ii) if $v_k^{\tau_j}$ is free in ψ , then $(h_j(Z))(k) = (h_j(Z'))(k)$, $j \neq i$, $j < q$.*

Returning now to the general case where A is either finite or infinite, we let $\text{Sat}_M(L', \mathfrak{A})$ be the set of those $\langle N, X, V, Z, u \rangle$ such that: (i) $\langle A, N, X, V \rangle$ is a model of λ ; and (ii) If f is an isomorphism of $\langle N, X \rangle$ onto $\langle \omega, Sc \rangle$ and if a_0 and h are determined in $\langle A, N, X, V \rangle$ by Δ_0 and $V(z, u, w)$ respectively, then $\langle Z, u \rangle \in \text{Sat}_I(L', \mathfrak{A}, f, a_0, h)$. Thus, for finite A , $\text{Sat}_M(L', \mathfrak{A})$ is empty. For infinite A we call $\text{Sat}_M(L', \mathfrak{A})$ the *modified satisfaction relation* (for L' in $\mathfrak{A}$).

We now begin the construction of an L_n -formula φ which implicitly defines this relation in terms of $R_0, \dots, R_{p-1}$. Vaught's construction on p. 296 of [2] will be followed closely. First we construct $4 + 2 \cdot q$ L'' -formulas whose only free variables are N, X and the individual variables indicated such that if $\mathfrak{M} = \langle A, N, X \rangle$, $N = \omega$, and $X = Sc$, then:

$\text{Fm}' = \text{Fm}'(u)$ holds for $(\mathfrak{M}, u)$ if and only if $u \in L'$;

$\text{Fm}'' = \text{Fm}''(u)$ holds for $(\mathfrak{M}, u)$ if and only if $u \in L''$;

$\text{Tf} = \text{Tf}(u, u')$ holds for $(\mathfrak{M}, u, u')$ if and only if $u \in L'$ and $u' = g(u)$, where g is a function satisfying the conditions of Lemma 2;

$\text{Cn}_2 = \text{Cn}_2(u, u', u'')$ holds for $(\mathfrak{M}, u, u', u'')$ if and only if u and u' are expressions (formed from variables, $/$, and $\exists$), and u'' is the expression formed by concatenating u with u' ;

$\text{Vb}_i = \text{Vb}_i(u, u')$ holds for $(\mathfrak{M}, u, u')$ if and only if $u \in \omega$ and $u' = v_u^{\tau_i}$, $i < q$;

$\text{Fr}_i = \text{Fr}_i(u, u')$ holds for $(\mathfrak{M}, u, u')$ if and only if $u \in \omega$, $u' \in L_n$, and $v_u^{\tau_i}$ occurs free in u' , $i < q$.

Let $\text{Cn}_{r+1}(v_0, \dots, v_{r-1}, v_r, v_{r+1}) = \exists v_{r+2} [\text{Cn}_r(v_0, \dots, v_{r-1}, v_{r+2}) \wedge \text{Cn}_2(v_{r+2}, v_r, v_{r+1})]$, $2 \leq r < \omega$. Let $\langle \vartheta_0, \dots, \vartheta_{q-1} \rangle$ be the $[n-1; \tau_0, \dots, \tau_{q-1}]$ -sequence of formulas. As before, let $\vartheta_i(v_j, v_k^{\tau_i})$ result from $\vartheta_i = \vartheta_i(u, W^{\tau_i})$ by substitution that has been preceded by required changes of bound

variables, and let $\vartheta_i(v_m^{(0)^{n-1}}, v_j)$ result from $\vartheta_i = \vartheta_i(Z, u)$ similarly, $i < q$. Let $Z' = v_3^{(0)^{n-1}}$ and for $i < q$ let $\text{Ef}_i = \text{Ef}_i(Z^{n-1}, Z', v_0, v_1)$ be a conjunction of the following q formulas:

$$\forall v_2[[\text{Fr}_i(v_2, v_0) \wedge \neg v_2 \cong v_1] \supset \forall W^{\tau_i}[\vartheta_i(Z^{n-1}, v_2) \equiv \vartheta_i(Z', v_2)]], \text{ and} \\ \forall v_2[\text{Fr}_j(v_2, v_0) \supset \forall W^{\tau_j}[\vartheta_j(Z^{n-1}, v_2) \equiv \vartheta_j(Z', v_2)]], j \neq i, j < q.$$

Finally, let S be a variable of type $((0), (0, 0), (0, 0, 0), (0)^{n-1}, 0)$ which is distinct from $R_0, \dots, R_{p-1}$.

Now let $1 \leq i < q$, so that $\tau_i = (\tau_{i_1}, \dots, \tau_{i_r})$ for some $\{\tau_{i_1}, \dots, \tau_{i_r}\} \subseteq \{\tau_0, \dots, \tau_{q-1}\}$. We now form a φ_i which roughly speaking describes when our satisfaction relation holds for atomic formulas $v_{j_0}^{\tau_{i_1}} v_{j_1}^{\tau_{i_2}} \dots v_{j_r}^{\tau_{i_r}}$. Let $W_i, W_{i_1}, \dots, W_{i_r}$ be $v_{2r+3}^{\tau_{i_1}}, v_{2r+4}^{\tau_{i_2}}, \dots, v_{3r+3}^{\tau_{i_r}}$ respectively. Then let φ_i be:

$$\forall Z^{n-1} \forall v_0 \dots \forall v_{2r+2} \forall W_i \forall W_{i_1} \dots \forall W_{i_r} [[\text{Vb}_i(v_0, v_1) \wedge \text{Vb}_{i_1}(v_2, v_3) \\ \wedge \dots \wedge \text{Vb}_{i_r}(v_{2r}, v_{2r+1}) \wedge \vartheta_i(v_0, W_i) \wedge \vartheta_{i_0}(v_2, W_{i_1}) \wedge \dots \wedge \\ \vartheta_{i_r}(v_{2r}, W_{i_r}) \wedge \text{Cn}_{r+1}(v_1, v_3, \dots, v_{2r+1}, v_{2r+2})] \supset \\ [S(N, X, V, Z^{n-1}, v_{2r+2}) \equiv W_i(W_{i_1}, \dots, W_{i_r})].$$

Next consider any R_i , $i < p$, and let its type be $(\tau_{i_1}, \dots, \tau_{i_r})$ so that $\{\tau_{i_1}, \dots, \tau_{i_r}\} \subseteq \{\tau_0, \dots, \tau_{q-1}\}$. We now form a φ_{q+i} corresponding similarly to the atomic formulas $R_i v_{j_1}^{\tau_{i_1}} \dots v_{j_r}^{\tau_{i_r}}$. Letting $W_i, \dots, W_{i_r}$ be $v_{2r+4}^{\tau_{i_1}}, \dots, v_{3r+3}^{\tau_{i_r}}$ respectively, we let φ_{q+i} be the formula:

$$\forall Z^{n-1} \forall v_1 \dots \forall v_{2r+2} \forall W_i \dots \forall W_{i_r} [[\Delta_{R_i}(v_1) \wedge \text{Vb}_i(v_2, v_3) \wedge \dots \wedge \\ \text{Vb}_{i_r}(v_{2r}, v_{2r+1}) \wedge \vartheta_{i_0}(v_2, W_{i_1}) \wedge \dots \wedge \vartheta_{i_r}(v_{2r}, W_{i_r}) \wedge \\ \text{Cn}_{r+1}(v_1, v_3, \dots, v_{2r+1}, v_{2r+2})] \supset [S(N, X, V, Z^{n-1}, v_{2r+2}) \equiv \\ R_i(W_{i_1}, \dots, W_{i_r})].$$

To describe when satisfaction holds for an L'' -formula ψ/ψ' , we let φ_{q+p+1} be the formula:

$$\forall Z^{n-1} \forall v_0 \dots \forall v_3 [[\Delta_{(v_0)} \wedge \text{Fm}''(v_1) \wedge \text{Fm}''(v_2) \wedge \text{Cn}_3(v_0, v_1, v_2, v_3)] \supset \\ [S(N, X, V, Z^{n-1}, v_3) \equiv [S(N, X, V, Z^{n-1}, v_1)/S(N, X, V, Z^{n-1}, v_2)]]].$$

To describe when satisfaction holds for an L'' -formula $\exists v_m^{\tau_i} \psi$, $i < q$, we let $Z' = v_3^{(0)^{n-1}}$ and let $\varphi_{q+p+1+i}$ be the formula:

$$\forall Z^{n-1} \forall v_0 \dots \forall v_4 [[\Delta_3(v_0) \wedge \text{Vb}_i(v_1, v_2) \text{Fm}''(v_3) \wedge \text{Cn}_3(v_0, v_2, v_3, v_4)] \supset \\ [S(N, X, V, Z^{n-1}, v_4) \equiv \exists Z' [\text{Ef}_i(Z^{n-1}, Z', v_3, v_1) \wedge S(N, X, V, Z', v_3)]]].$$

Turning from clauses for L'' -formulas to clauses for others, we let φ_{2q+p+2} and φ_{2q+p+3} be respectively:

$$\forall Z^{n-1} \forall v_0 \forall v_1 [[\text{Fm}'(v_0) \wedge \text{Tf}(v_0, v_1)] \supset [S(N, X, V, Z^{n-1}, v_0) \equiv \\ S(N, X, V, Z^{n-1}, v_1)]], \text{ and}$$

$$\forall Z^{n-1} \forall v_0 [\neg \text{Fm}'(v_0) \supset \neg S(N, X, V, Z^{n-1}, v_0)].$$

Finally we let φ be the formula:

$$\forall N \forall X \forall V [\neg \lambda \supset \forall Z^{n-1} \forall u \neg S(N, X, V, Z^{n-1}, u)] \wedge [\lambda \supset \varphi_1 \wedge \dots \wedge \varphi_{2q+p+3}].$$

From lemmas 1 and 3, the construction of φ , and the definition of Sat_M there follows:

THEOREM 1. *We can construct an L_n -formula φ whose only free variables are $R_0, \dots, R_{p-1}$ and S of types $\sigma_0, \dots, \sigma_{p-1}$ and $((0), (0, 0), (0, 0, 0), (0)^{n-1}, 0)$ respectively such that if $\mathfrak{A} = \langle A, R_0, \dots, R_{p-1} \rangle$ and $R_0 \in \sigma_0 A, \dots, R_{p-1} \in \sigma_{p-1} A$, then $(\mathfrak{A}, S)$ is a model of φ if and only if $S = Sat_M(L', \mathfrak{A})$.*

We now consider briefly the more general case where $R_0, \dots, R_{p-1}$ are of order $\leq n$. Let $\tau_0, \dots, \tau_{q-1}$ be as before. If each of $R_0, \dots, R_{p-1}$ is of a type different from $\tau_0, \dots, \tau_{q-1}$ then we construct φ as before. Assume now, for example, that $R_i = v_k^j$, $i < p$, $j < q$, $k < \omega$. Then for $\mathfrak{A} = \langle A, R_0, \dots, R_{p-1} \rangle$ we consider only those $\langle e_0, \dots, e_{q-1} \rangle$ such that $e_j(k) = R_i$, and define $Sat_I(L', \mathfrak{A}, f, a_0, h)$ accordingly. It is convenient to define the relation first for those L'' -formulas in which there is no bound occurrence of R_i and only afterwards for the remaining L'' -formulas. In constructing φ we then everywhere relativize Z^{n-1} in a manner corresponding to the condition $e_j(k) = R_i$. We shall not give details.

In conclusion, note that we can simplify if we restrict ourselves to the natural numbers. More precisely, let $A = \omega$, $R_0 = Sc$, f the identity map of ω onto ω , $a_0 = 0$, and h some mapping of A onto $A^{(\omega)}$ determined in $\langle A, Sc \rangle$ by an L_2 -formula with the free variables R_0, z, u, w in a sense which should be obvious. Then, proceeding as in the proof of Theorem 1, we can construct for any $R_1, \dots, R_{p-1}$ of order $\leq n$ an L_n -formula φ^* whose only free variables are $R_0, R_1, \dots, R_{p-1}$ and S , where S is of type $((0)^{n-1}, 0)$, such that if $R_1, \dots, R_{p-1}$ are of a type appropriate for $R_1, \dots, R_{p-1}$ and if $\mathfrak{A} = \langle A, R_0, R_1, \dots, R_{p-1} \rangle$, then $(\mathfrak{A}, S)$ is a model of φ^* if and only if $S = Sat_I(L', \mathfrak{A}, f, a_0, h)$.

§ 4. Consequences. We now turn to the problem of showing that for infinite $\mathfrak{A}$ no L' -formula defines $Sat_M(L', \mathfrak{A})$ explicitly in terms of $R_0, \dots, R_{p-1}$. We first state a lemma for L' whose first order analogue amounts to a certain relativization of Theorem 1, p. 46, of [6]. To achieve similarity of nomenclature with II.2 of [6], we let a *sentence* be an L' -formula in which no variable other than $R_0, \dots, R_{p-1}$ occurs free, and we call a set W of sentences which contains each sentence that holds for every model of W the set of sentences *valid* in the *theory* $T = T(W)$ determined by W . If, and only if, no sentence and its negation are both in W , then T is *consistent*. For any set $P \subseteq \omega$ and any π we say that π *defines* P in T (relative to the formulas λ and $\Delta_0, \Delta_1, \dots$) if and only if π is an L' -formula whose free variables are $R_0, \dots, R_{p-1}, N, X, V, u$ such that, for each $j \in \omega$, $\forall N \forall X \forall V [\lambda \supset \forall u [\Delta_j \supset \pi]]$ or $\forall N \forall X \forall V [\lambda \supset \forall u [\Delta_j \supset$

$\neg\pi$] is valid in T depending on whether $j \in P$ or $j \notin P$. For any function f from ω into ω and any ξ we say that ξ defines f in T (relative to λ and $\Delta_0, \Delta_1, \dots$) if and only if ξ is an L' -formula whose free variables are $R_0, \dots, R_{p-1}, N, X, V, u, u'$ such that for each $j, k \in \omega$:

- (i) If $f(j)=k$, then $\forall N \forall X \forall V [\lambda \supset \forall u \forall u' [\Delta_j \wedge \Delta_k(u') \supset \xi]]$ is valid in T ;
- (ii) If $f(j) \neq k$, then $\forall N \forall X \forall V [\lambda \supset \forall u \forall u' [\Delta_j \wedge \Delta_k(u') \supset \neg \xi]]$ is valid in T ; and
- (iii) $\forall N \forall X \forall V [\lambda \supset \forall u \forall u' \forall u'' [\Delta_j \wedge \xi(u') \wedge \xi(u'') \supset u' \simeq u'']]$, is valid in T , where $\xi = \xi(u')$.

A diagonal function (relative to λ and $\Delta_0, \Delta_1, \dots$) shall be any function f from ω into ω such that if $j = \forall N \forall X \forall V [\lambda \supset \psi]$ and ψ is an L' -formula then $f(j) = \forall N \forall X \forall V [\lambda \supset \forall u [\Delta_j \supset \psi]]$.

As on p. 47 of [6] one can then prove:

LEMMA 5. Let f be a diagonal function and let W be the set of valid sentences of a theory T , let T be consistent, and let $\exists N \exists X \exists V \lambda$ and $\forall N \forall X \forall V [\lambda \supset \exists u \Delta_j]$, $j = 0, 1, \dots$ be valid in T . Then f and W are not both definable in T .

THEOREM 2. Consider any $\mathfrak{A} = \langle A, R_0, \dots, R_{p-1} \rangle$ such that A is infinite and $R_0 \in \sigma_0 A, \dots, R_{p-1} \in \sigma_{p-1} A$. Then for the unique S such that $(\mathfrak{A}, S)$ is a model of the formula φ of Theorem 1 there is no L' -formula χ containing free only $R_0, \dots, R_{p-1}, N, X, V, Z^{n-1}$, and u , such that $(\mathfrak{A}, S)$ is a model of $\forall N \dots \forall u [S(N, X, V, Z^{n-1}, u) \equiv \chi]$.

PROOF. Let $\mathfrak{A}$ and S be as stated. Let W be the set of sentences having $\mathfrak{A}$ as a model. Then every sentence holding for every model of W is in W , so that W is the set of valid sentences of a theory T . Evidently, T is consistent, $\exists N \exists X \exists V \lambda$ and $\forall N \forall X \forall V [\lambda \supset \exists u \Delta_j]$, $j = 0, 1, \dots$ are valid in T , and some diagonal function is definable in T . There is an L' -formula Sent containing free only N, X , and u such that if $\langle A', N, X \rangle = \langle A', \omega, Sc \rangle$, then $\langle A', N, X, u \rangle$ is a model of Sent if and only if u is a sentence. Now suppose there were an L' -formula χ containing free only $R_0, \dots, u$ such that $(\mathfrak{A}, S)$ is a model of $\forall N \dots \forall u [S(N, X, V, Z^{n-1}, u) \equiv \chi]$. Then by Lemma 3 the L' -formula Sent $\wedge \exists Z^{n-1} \chi$ would define W in T , thus contradicting Lemma 5. Q.E.D.

Beth's theorem, somewhat loosely stated, asserts that each first order theory T whose nonlogical constants are $R_0, \dots, R_{p-1}$ has the following property: If a first order formula φ contains a constant S in addition to $R_0, \dots, R_{p-1}$ and if for every model $\mathfrak{A}$ of T there is at most one S such that $(\mathfrak{A}, S)$ is a model of φ , then there is a formula χ of T which explicitly defines S in terms of $R_0, \dots, R_{p-1}$ for every such $(\mathfrak{A}, S)$. From Theorems 1 and 2 it follows that each T which is an n -th order theory in our sense, $2 \leq n < \omega$, which has only finitely many nonlogical constants, and which has at least one infinite model lacks the analogous property. Moreover,

given $R_0, \dots, R_{p-1}$ and n , the same formula φ can be used to account for the lack in each case. Furthermore, for this φ there is for each model of T at least one S such that $(\mathfrak{A}, S)$ is a model of φ . Hence replacement in the hypothesis of Beth's Theorem of 'at most' by 'exactly' is of no avail.

The case where $p = 0$ deserves special discussion. One may ask, somewhat vaguely, what *logical* notions S can be defined by using bound variables of order $< n$ only. For simplicity, we confine ourselves to a fixed infinite universe A .² Then the question seems to concern those sets $S \in \tau A$ such that $o(\tau) \leq n$ and such that S is *invariant* (on A) in the following sense: For any 1-1-mapping π of A onto A , if π_* is the induced 1-1-mapping of τA onto τA , then $\pi_*(S) = S$. Now there are no invariant S of order 0 on A , and each invariant S of order 1 on A can evidently be defined by using only individual variables, truth functional connectives, and a symbol for identity. Consider therefore the case where $2 \leq n < \omega$. Then there are at least two ways of making the initial question precise (cf. [5], p. 108). First, let $\chi \in L^* \subset L_n$ if and only if each variable free in χ is of order $< n$, and let χ *define explicitly* (on A) the set S of those $\langle X_0, \dots, X_{m-1} \rangle$ such that $\langle A, X_0, \dots, X_{m-1} \rangle$ is a model of χ . Second, let an L_n -formula φ *define implicitly* (on A) a set S if and only if φ contains a single free variable, $\langle A, S \rangle$ is a model of φ , and if $S' \neq S$ then $\langle A, S' \rangle$ is not a model of φ . In either case the S defined is invariant. Theorems 1 and 2 show that $\text{Sat}_M(L^*, A)$ can be defined implicitly using bound variables of order $< n$ only but not explicitly. This indicates a certain expressive inadequacy of L^* .

The theory T in the proof of Theorem 2 is complete in the sense that for every sentence ψ , either ψ or $\neg\psi$ is valid in T . It follows that there are no L' -formulas $\chi_0, \dots, \chi_{k-1}$ containing free only $R_0, \dots, u$ such that the $(\mathfrak{A}, S)$ of Theorem 2 is a model of $\forall N \dots \forall u [S(N, \dots, u) \equiv \chi_0] \vee \dots \vee \forall N \dots \forall u [S(N, \dots, u) \equiv \chi_{k-1}]$. Hence, for $2 \leq n < \omega$, there is also no obvious analogue of the theorem of Svenonius [4].

The interpolation theorem for first order formulas which concerns predicate symbols also has no analogue for L_n -formulas, $2 \leq n < \omega$. For such an analogue would yield an analogue of Beth's Theorem (cf. [3]). For $p = 0$ this implies that even for those formulas $\phi \supset \phi'$ true for all appropriate $\mathfrak{A}$ such that ϕ and ϕ' have no free variable in common, interpolation is not always possible.

The discussion on pp. 306, 307 of [2] strongly suggests that a result similar to Theorem 2.1 of [2] holds for higher order languages in place of first order languages, with definability in an appropriate higher order number theory in place of axiomatizability or recursiveness as a sufficient condition.

² For finite A , each invariant $S \in \tau(A)$ can be explicitly defined using bound variables of order $< n$ only.

Theorem 1 allows us to slightly strengthen the result there suggested. Let ξ define a set $M \subseteq \omega$ in n -th order number theory, $2 \leq n < \omega$, if and only if ξ is an L_n -formula (i.e., a formula whose bound variables are of order $< n$) whose only free variables are X and u such that if $\langle N, X \rangle = \langle \omega, Sc \rangle$ then $\langle N, X, u \rangle$ is a model of ξ if and only if $u \in M$.

THEOREM 3. Let $2 \leq n < \omega$, let M be a set of L_n -formulas which contain free only $R_0, \dots, R_{p-1}$, let M be definable in n -th order number theory, and let φ be the L_n -formula of Theorem 1. Then there is an L_n -formula μ containing free only $R_0, \dots, R_{p-1}$ and S such that $M, \exists! S[\varphi \wedge \mu], \exists S[\varphi \wedge \mu]$, and $\forall S[\varphi \supset \mu]$ have the same infinite models.

PROOF. Let M and φ be as stated and let ξ define M in n -th order number theory. Let $\xi^{(N)}$ result from ξ by relativization of individual quantifiers to N . Let μ be

$$\forall N \forall X \forall V \forall Z^{n-1} [\exists u S(N, X, V, Z^{n-1}, u) \supset \forall u [\xi^{(N)} \supset S(N, X, V, Z^{n-1}, u)]].$$

One can then verify that μ satisfies the conclusion of the theorem.

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